Do Households Substitute Intertemporally? Ten structural shocks that suggest not.

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Federal Reserve Board

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Section 1

Introduction

How does Consumption change with Interest Rates?

Problem: Interest rates are, by their nature, general equilibrium outcomes.

- ‘Idiosyncratic’ interest rate shocks are exceedingly rare.
- Unlike in the MPC literature, micro data doesn’t help us much.

How does Consumption change with Interest Rates?

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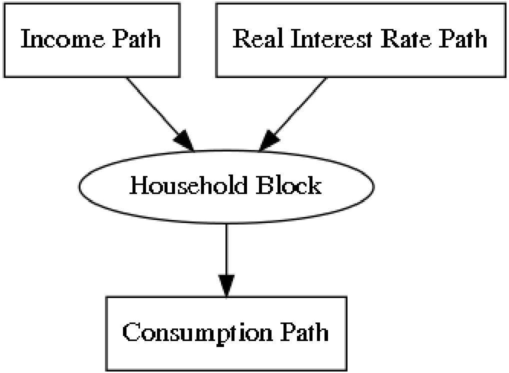
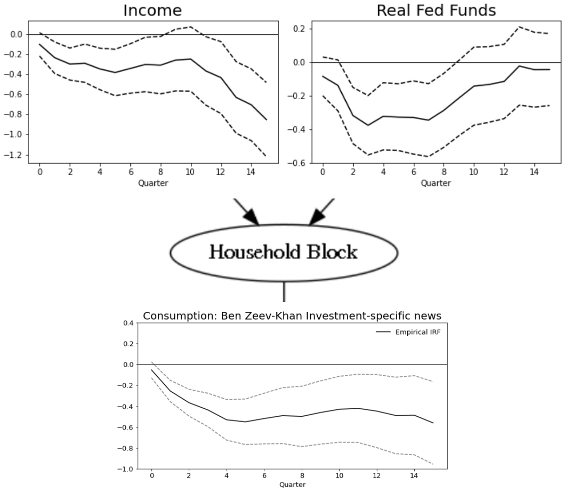
My Solution:

- Find macro shocks that should affect consumption only indirectly through aggregate income and interest rates.
  - e.g. Monetary policy shocks, fiscal shocks, technology shocks.
  - Ramey [2016] provides ten of these.
- Estimate the consumption response to income from micro data.
- The consumption response to interest rates is the residual after accounting for income.

Sequence Space view of the Household



Example: Ben Zeev-Khan Investment-specific News



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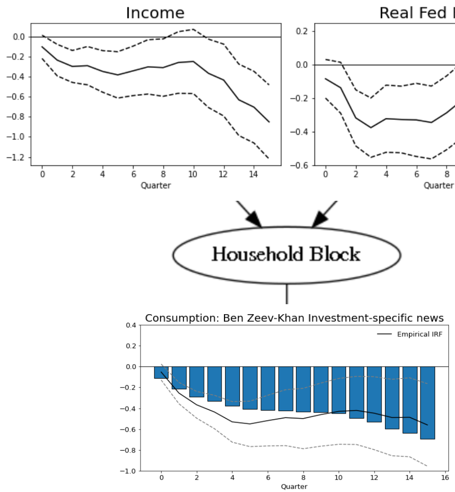
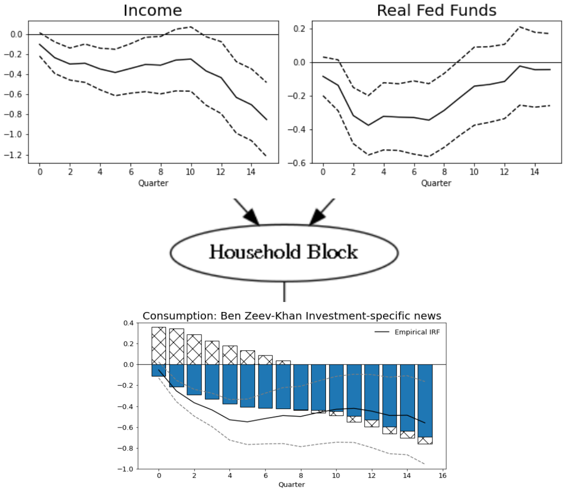


Example: Ben Zeev-Khan Investment-specific News



Very Brief Literature Review on Intertemporal Substitution

Hall [1988] - seminal paper

Euler equation estimation reviewed in Browning and Lusardi [1996]

- But "Death to the Log-Linearized Euler Equation," Carroll [1997]

Some recent papers disagree:

- Best et al. [2019]
- Crump et al. [2022]

Section 2

Methodology

A Toy Endowment Economy Model to Set Ideas

Two Agents

- Hand-to-mouth fraction λ

$$C_{htm,t} = Y_t$$

- Optimizing fraction $1 - \lambda$

$$\underset{\{C_{opt,t}\}}{\text{maximize}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{C_{opt,t}^{1-\sigma}}{1-\sigma}$$

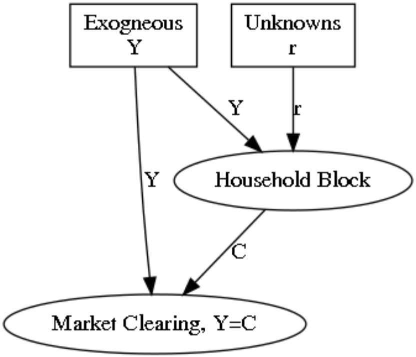
subject to:

$$C_{opt,t} + A_t \leq (1 + r_t)A_{t-1} + Y_t$$

Y_t is exogenous.
r_t is endogenous and clears the market.

A Toy Endowment Economy Model to Set Ideas

A Directed Acyclical Graph Representation



r_t is endogenous and clears markets:

- $C_t = \lambda C_{htm,t} + (1 - \lambda) C_{opt,t} = Y_t$
- $A_t = 0$

A Sequence Space Representation

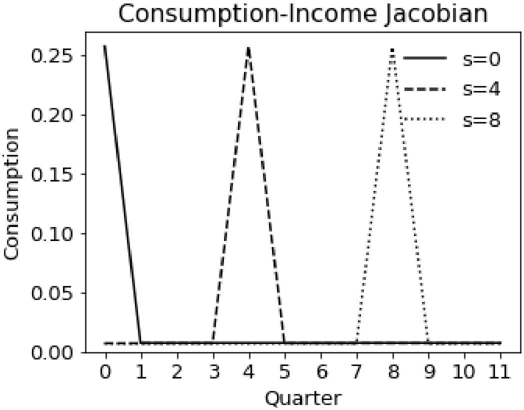
The Aggregate Consumption Function:

$$\{C\}_{t=0}^{\infty} = \mathcal{C}(\{Y\}_{t=0}^{\infty}, \{r\}_{t=0}^{\infty})$$

To study deviations from the steady state, two Jacobians are sufficient statistics:

- $\frac{d\mathcal{C}_t}{dY_s}$ Consumption-to-income
  (the intertemporal MPC)
  We take as given (in this case from the model, in practice from data)
- $\frac{d\mathcal{C}_t}{dr_s}$ Consumption-to-real rates
  (our object of interest)

The Toy Model Consumption-to-Income Jacobian

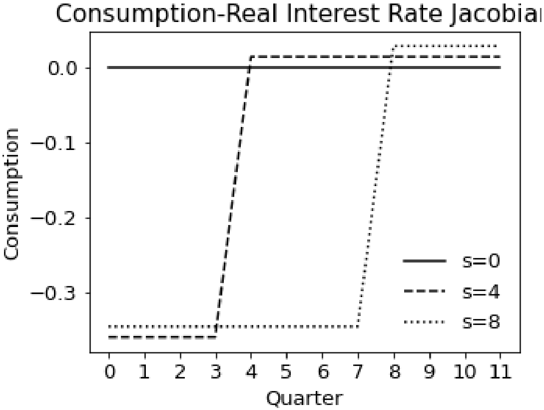


$$\frac{dC_t}{dY_s}$$

Also known as the Intertemporal Marginal Propensity to Consume.

Assume we know this from microdata.

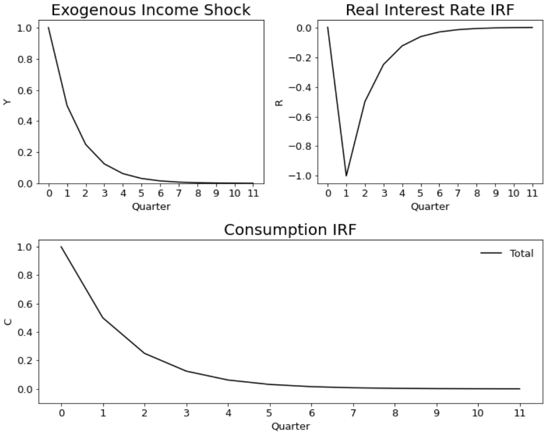
The Toy Model Consumption-to-Real Interest Rates Jacobian



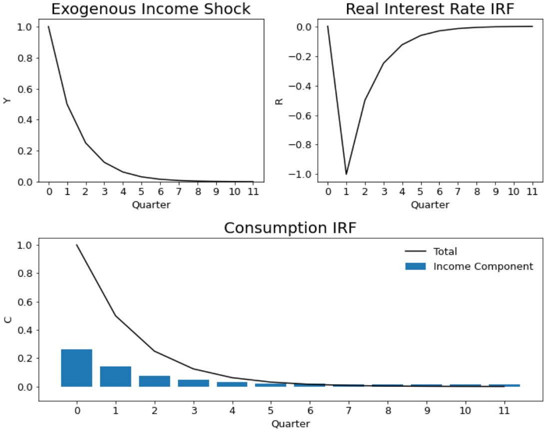
$$\frac{d\mathcal{C}_t}{dR_s}$$

Assume we know the *shape* but not the *magnitude* of this Jacobian.

The Toy Model Income Shock



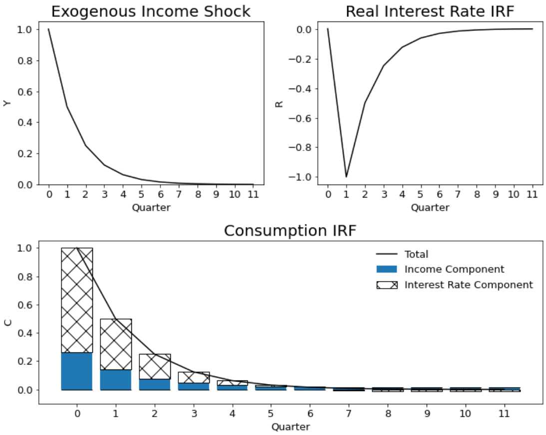
The Toy Model Income Shock



Income Component

$$= \sum_{s=0}^{\infty} \frac{d\mathcal{C}_t}{dY_s} \Delta Y_s$$

The Toy Model Income Shock



Income Component

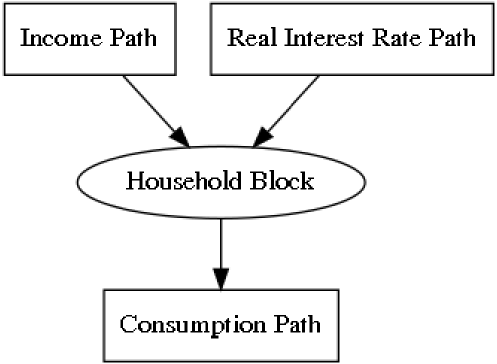
$$= \sum_{s=0}^{\infty} \frac{dC_t}{dY_s} \Delta Y_s$$

Real Interest Rate Component

$$= \sum_{s=0}^{\infty} \frac{dC_t}{dR_s} \Delta R_s$$

We can recover the *magnitude* of $\frac{dC_t}{dR_s}$ required to match the consumption IRF.

Bringing the Methodology to the Data

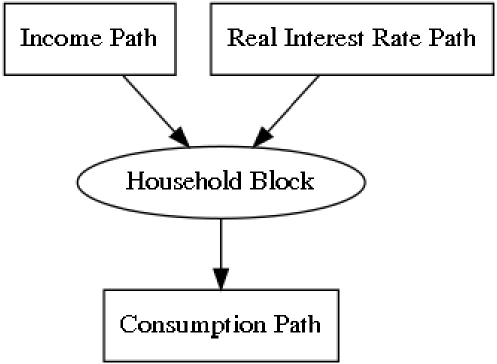


Model bells and whistles often keep the input-output structure of the Household Block unchanged:

- Financial frictions
- Sticky prices and wages
- Time-to-build
- Heterogenous households
- Heterogenous firms
- International Trade

I will also include equity and house prices as inputs.

Bringing the Methodology to the Data



No matter how complicated the rest of the model, this input-output structure suggests sufficient statistics for the Household Block are:

$$\frac{dC_t}{dY_s} \text{ and } \frac{dC_t}{dR_s}$$

(and Jacobians for equity and house prices).

I will use a model to reduce the dimensionality of these Jacobians and then estimate the model using both micro and macro data.

A minimal model that can capture the empirical evidence on intertemporal MPC is a one-asset heterogenous agent model.

Baseline model: One asset heterogenous-agent model

Off-the-shelf one asset heterogenous-agent model

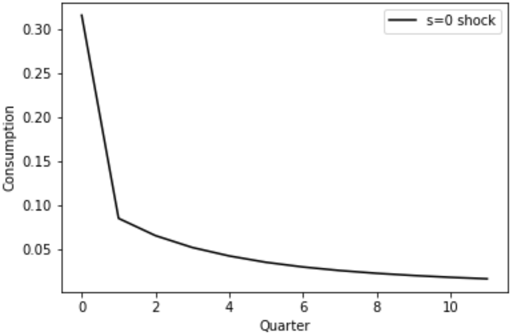
$$V(e,a) = \max_{c,a'} u(c) + \beta \mathbb{E}[V(e',a')|e]$$
$$\text{s.t. } a' + c = (1+R)a + y(e)$$
$$a' \geq 0$$

Parameterization:

- CRRA = 1
- R = 3%
- Income process from Kaplan & Violante (2018) with progressive taxes

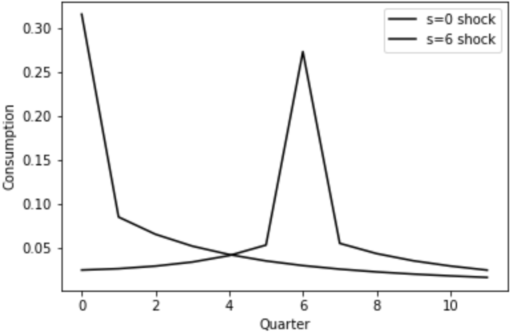
Population split into six groups with different β's to match empirical iMPCs.

The Income Jacobian $\frac{d\mathcal{C}_t}{dY_s}$



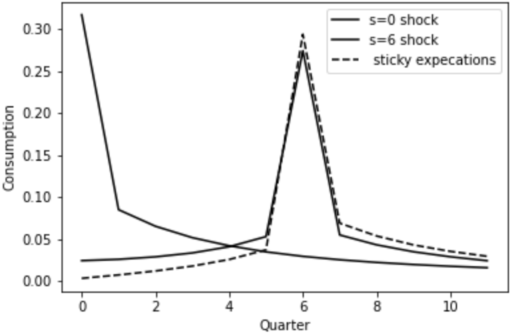
The first column, $\frac{d\mathcal{C}_t}{dY_0}$ is is calibrated to match Fagereng et al. [2021]

The Income Jacobian $\frac{d\mathcal{C}_t}{dY_s}$



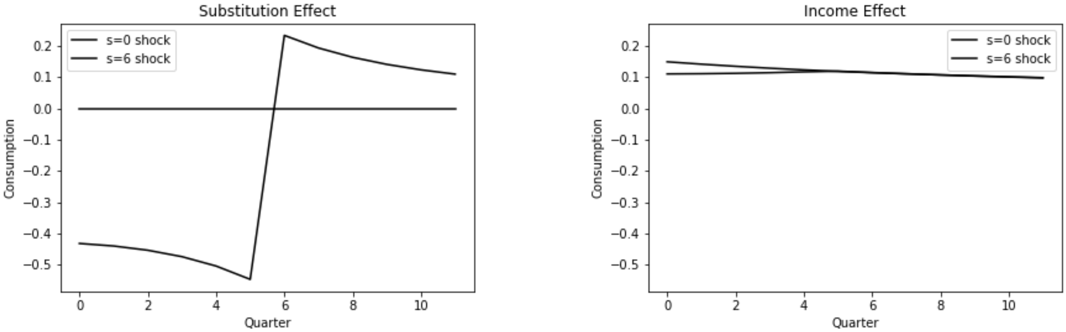
Less is known about anticipated income shocks (later columns of $\frac{d\mathcal{C}_t}{dY_s}$)
Model has too much foresight? e.g. Ganong and Noel [2019], Kueng [2018], Parker [1999]

The Income Jacobian $\frac{dC_t}{dY_s}$



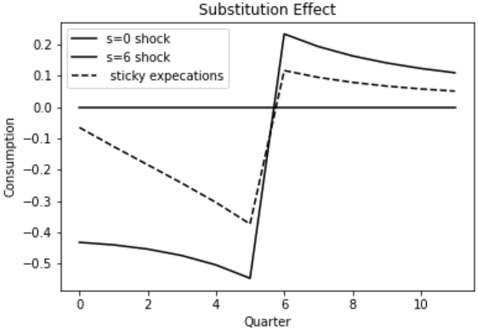
One fix is "sticky expectations". Each quarter a fraction θ_{inc} of households become attentive.
Carroll et al. [2020], Auclert et al. [2020]
The income jacobian is calibrated to microdata, except θ_{inc} which is estimated with macro IRFs.

The Interest Rate Jacobian $\frac{dC_t}{dR_s}$



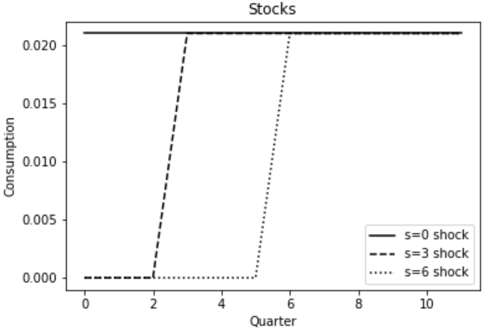
$\frac{dC_t}{dR_s}$ can be divided into an income and substitution effect—Farhi et al. [2022].
I replace the income effect with an asset-by-asset approach (described later).

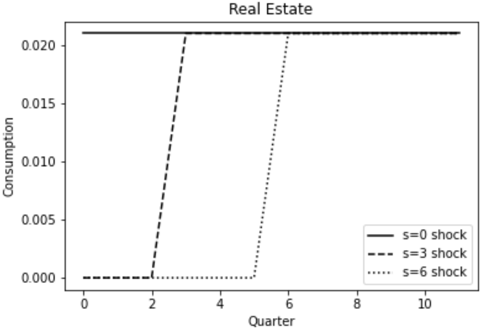
The Interest Rate Jacobian $\frac{dC_t}{dR_s}$



- I apply sticky expectations to the substitution effect.
- Separate parameter θ_{sub} estimated with macro IRFs.
- Preview: I estimate $\theta_{sub} = 0$, there is no substitution effect.

Stocks and Real Estate





I replace the income component of the interest rate jacobian with jacobians for stocks and real estate.
Consistent with micro literature: Annual MPC of 0.03 from wealth.
Results are insensitive to exact specification.

Taking Stock

Using a mixture of microdata and theory, two parameters span the space of reasonable Jacobians:

Income Jacobian

- $\theta_{inc} = 1$: Full anticipation of future income.
- $\theta_{inc} = 0$: No anticipation, but still a large consumption response when income arrives.

Intertemporal Substitution Jacobian:

- $\theta_{sub} = 1$: Full intertemporal substitution.
- $\theta_{sub} = 0$: No intertemporal substitution at all.

Plan: Find values of θ_{inc} and θ_{sub} that best fit identified structural shocks.

Empirical IRF Estimation

I run Jorda projections for outcome O on shock series ϵ^q:

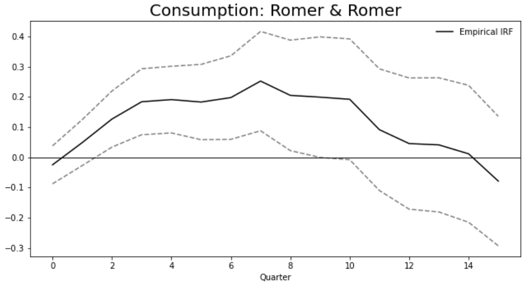
$$O^q_{t+h} = J^{O,q}_h \epsilon^q_t + \beta^{O,q}_h X_t + \varepsilon^{O,q}_{t,h}$$

to get estimates of the impulse response $\hat{J}^{O,q}_h$ along with standard errors.

O contains the output IRF C as well as all input IRFs $Y, R, \text{stocks}, \text{realestate}$.

For example:

- q is Romer and Romer [2004] shock series.
- O is consumption.
- $J^{O,q}$, the IRF of consumption to a Romer and Romer [2004] shock, is plotted on the right.



Jacobian Estimation

Given parameters $\boldsymbol{\theta} = (\theta_{inc}, \theta_{sub})$, the consumption Jacobian for input O is

$$\frac{d\mathcal{C}_t}{dO_s}(\boldsymbol{\theta}).$$

Using the estimated input IRFs $\hat{\boldsymbol{J}}^{\boldsymbol{q}} = (\hat{J}^{Y,q}, \hat{J}^{R,q}, \hat{J}^{\text{stocks},q}, \hat{J}^{\text{realestate},q})$, I calculate the implied consumption IRF:

$$\mathcal{C}(\hat{\boldsymbol{J}}^{\boldsymbol{q}}, \boldsymbol{\theta}) = \sum_{O \in \{Y, R, \text{stocks}, \text{realestate}\}} \frac{d\mathcal{C}_t}{dO_s}(\boldsymbol{\theta}) \hat{J}^{Y,q}$$

which I will compare the empirical consumption IRF $\hat{\mathcal{C}}^q = \hat{J}^{C,q}$

Jacobian Estimation

For each shock series, I calculate an objective function given $\boldsymbol{\theta}$:

$$V^q(\boldsymbol{\theta}) = (\mathcal{C}(\hat{\boldsymbol{J}}^{\boldsymbol{q}}, \boldsymbol{\theta}) - \hat{\mathcal{C}}^q)'\Sigma_q^{-1}(\mathcal{C}(\hat{\boldsymbol{J}}^{\boldsymbol{q}}, \boldsymbol{\theta}) - \hat{\mathcal{C}}^q)$$

where Σ_q is a diagonal matrix of consumption IRF variances.

For a set Q of shock series, I estimate $\hat{\boldsymbol{\theta}}$ as

$$\hat{\boldsymbol{\theta}} = \underset{\theta}{\text{argmin}} \sum_{q \in Q} V^q(\boldsymbol{\theta})$$

Choice of Structural Shock Series

Theory: Any structural shock that effects consumption only through aggregate changes to income, interest rates, stocks, and real estate

- Monetary policy shocks?
- TFP shocks?
- Government spending shocks?

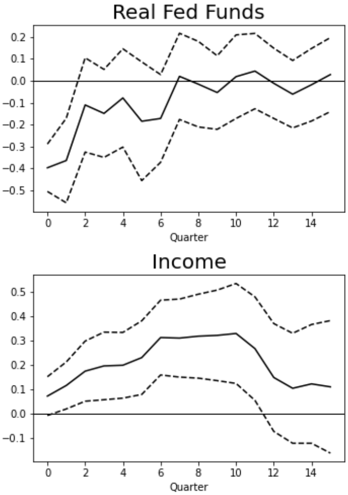
Problematic shocks might include taxes on the rich, oil shocks, pandemics.

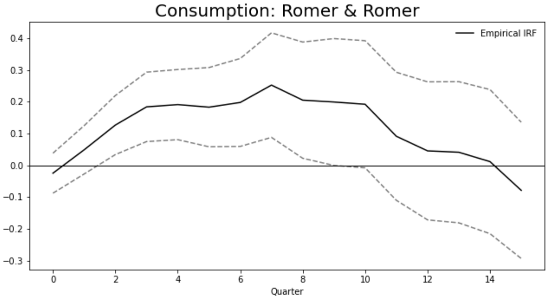
Practice: Take all ten shock series from Ramey [2016] used in Jorda projections that are not derived from a DSGE model.

Section 3

Results

Example 1: Romer & Romer





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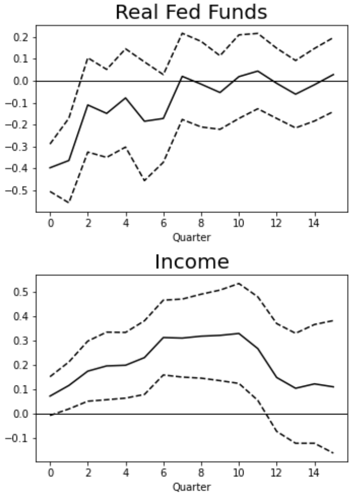
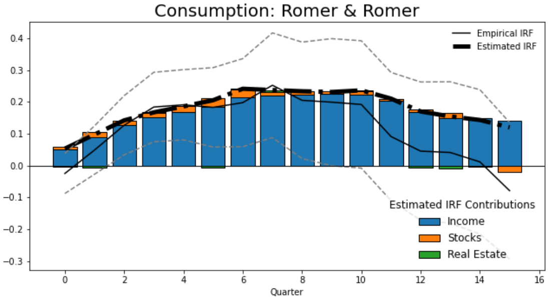
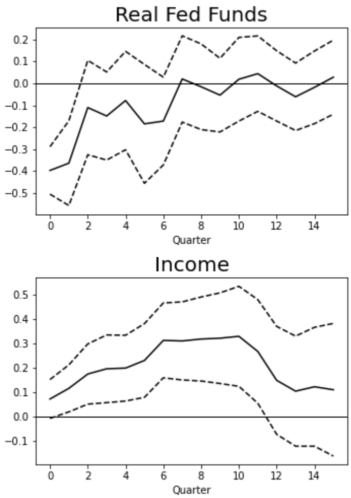
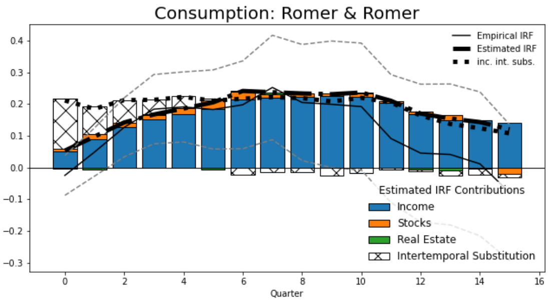




Income alone can explain the consumption path

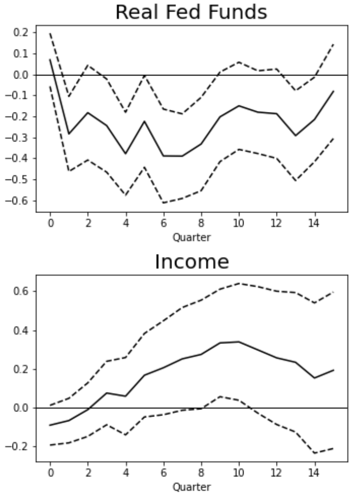
Example 1: Romer & Romer

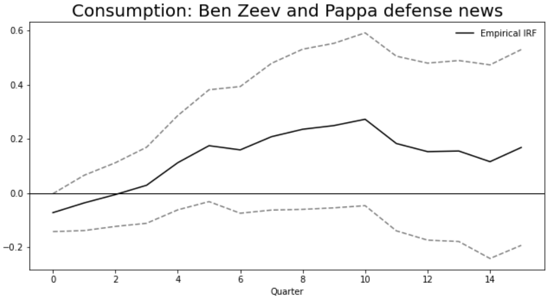




Intertemporal substitution worsens the fit

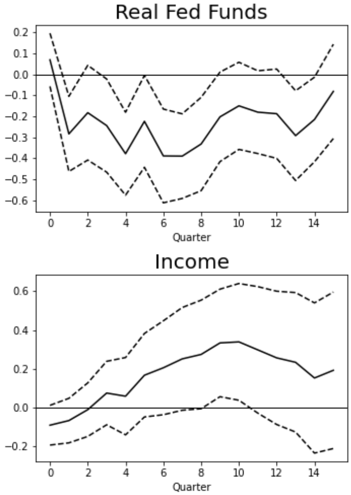
Example 2: Ben Zeev and Pappa defense news

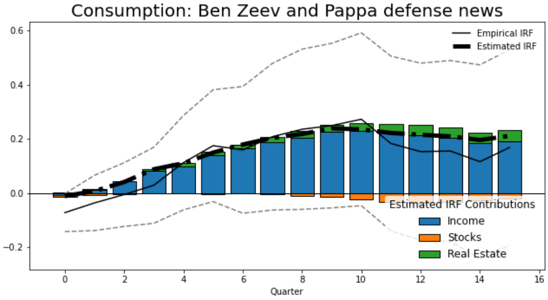




Real fed funds rate decreases **persistently**

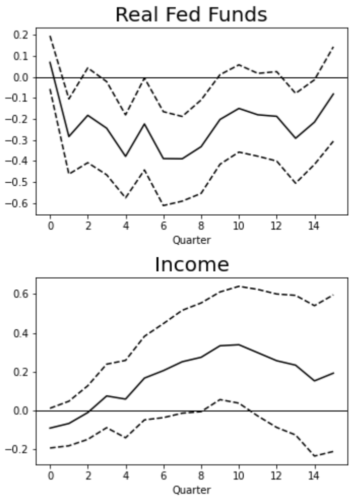
Example 2: Ben Zeev and Pappa defense news

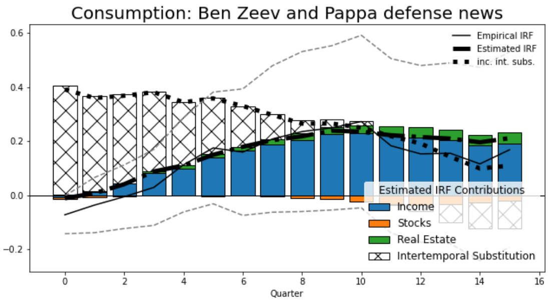




Again alone can explain the consumption path

Example 2: Ben Zeev and Pappa defense news





Again intertemporal substitution worsens the fit

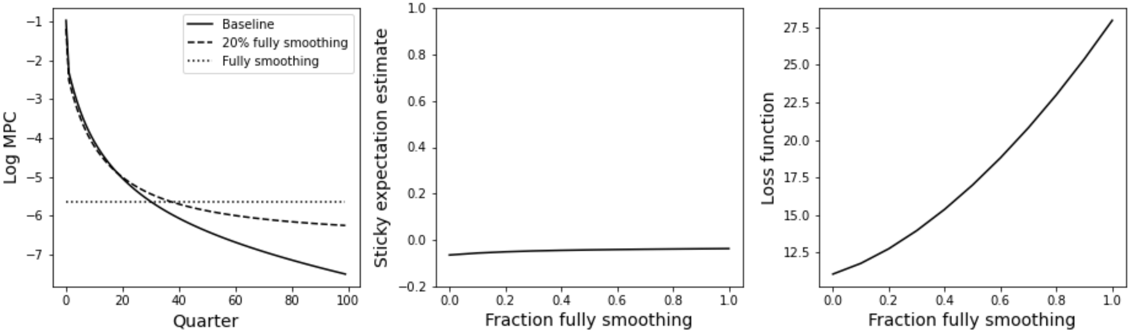
Parameter Estimates

Set of Structural Shocks (Q)	$\hat{\theta}_{sub}$		$\hat{\theta}_{inc}$	
All	**-0.04**	(0.05)	**0.26**	(0.19)
Romer and Romer [2004]	**-0.80**	(0.79)	**0.76**	(1.19)
Gertler and Karadi [2015]	**-0.13**	(1.17)	**0.00**	(0.25)
Ramey and Zubairy [2018] military news	**0.05**	(0.19)	**0.00**	(0.15)
Ben Zeev and Pappa [2017] defence spending shocks	**-0.01**	(0.10)	**0.11**	(0.43)
Mertens and Ravn [2012] tax news shocks	**0.05**	(0.13)	**0.01**	(0.96)
Leeper et al. [2012] expected taxes from one to five years forward	**-0.00**	(0.10)	**1.00**	(5.77)
Ben Zeev and Khan [2015] investment specific news shocks	**-0.14**	(0.18)	**0.09**	(0.13)
Fernald [2012] utilization-adjusted TFP	**-0.36**	(1.19)	**0.00**	(0.48)
Fernald [2012] utilization-adjusted investment TFP	**-0.68**	(0.95)	**0.00**	(1.47)
Francis et al. [2014] unanticipated TFP shocks	**-0.24**	(3.92)	**0.78**	(1.00)

Section 4

Robustness

Robustness to Lower MPCs



Robustness to model EIS assumption

Estimates of θ_{sub} for different model values of the elasticity of intertemporal substitution.

Set of Structural Shocks (Q)	EIS=1		EIS=0.5		EIS=0.33		EIS=0.25	
All	**-0.04**	(0.05)	**-0.10**	(0.09)	**-0.20**	(0.27)	**-0.33**	(0.46)
Romer and Romer [2004]	**-0.80**	(0.79)	**-1.00**	(1.61)	**-1.00**	(2.44)	**-1.00**	(3.16)
Gertler and Karadi [2015]	**-0.13**	(1.17)	**-0.38**	(7.73)	**-0.42**	(12.68)	**-0.42**	(17.18)
Ramey and Zubairy [2018]	**0.05**	(0.19)	**-0.68**	(2.03)	**-0.55**	(2.63)	**-0.52**	(3.16)
Ben Zeev and Pappa [2017]	**-0.01**	(0.10)	**-0.02**	(0.20)	**-0.04**	(0.34)	**-0.07**	(0.47)
Mertens and Ravn [2012]	**0.05**	(0.13)	**0.14**	(0.41)	**0.43**	(1.48)	**0.52**	(2.17)
Leeper et al. [2012]	**-0.00**	(0.10)	**-0.01**	(0.20)	**-0.02**	(0.32)	**-0.04**	(0.45)
Ben Zeev and Khan [2015]	**-0.14**	(0.18)	**-0.17**	(0.40)	**-0.17**	(0.60)	**-0.19**	(0.81)
Fernald [2012]	**-0.36**	(1.19)	**-0.75**	(3.24)	**-0.96**	(5.10)	**-1.00**	(6.57)
Fernald [2012]	**-0.68**	(0.95)	**-1.00**	(2.09)	**-1.00**	(3.14)	**-1.00**	(4.03)
Francis et al. [2014]	**-0.24**	(3.92)	**-0.30**	(13.38)	**-0.28**	(17.09)	**-0.24**	(15.89)

Finite Horizon Planning Estimates

Set of Structural Shocks (Q)	$\hat{\text{FHP}}_{sub}$		$\hat{\text{FHP}}_{inc}$	
All	**-0.53**	(0.18)	**0.87**	(0.12)
Romer and Romer [2004]	**-0.56**	(0.67)	**0.80**	(0.49)
Gertler and Karadi [2015]	**-0.62**	(1.23)	**0.14**	(8.61)
Ramey and Zubairy [2018] military news	**-0.26**	(2.03)	**0.83**	(0.51)
Ben Zeev and Pappa [2017] defence spending shocks	**-0.31**	(0.80)	**0.83**	(0.58)
Mertens and Ravn [2012] tax news shocks	**0.64**	(0.38)	**-0.08**	(11.35)
Leeper et al. [2012] expected taxes from one to five years forward	**-0.22**	(1.21)	**1.00**	(0.32)
Ben Zeev and Khan [2015] investment specific news shocks	**-0.46**	(0.49)	**0.89**	(0.13)
Fernald [2012] utilization-adjusted TFP	**-0.70**	(0.81)	**-0.16**	(11.20)
Fernald [2012] utilization-adjusted investment TFP	**-0.92**	(0.29)	**-0.27**	(70.02)
Francis et al. [2014] unanticipated TFP shocks	**-0.75**	(1.53)	**0.99**	(0.10)

Aside: Consumption Models are not Robust to Small Frictions

First order ‘mistakes’ in intertemporal substitution have second order welfare effects.

Cochrane [1989]:

- The cost of using a 1,5, or 10 year moving average in place of the actual interest rate is between 8 cents and $1.45 per quarter.
- “Theory as it stands provides few predictions about the relationship between aggregate consumption and asset price or aggregate quantity fluctuations that are robust to $1 “mistakes” or misspecifications.”

Theory leaves us with wide priors

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Cochrane [1989]:

- The cost of using a 1,5, or 10 year moving average in place of the actual interest rate is between 8 cents and $1.45 per quarter.
- “Theory as it stands provides few predictions about the relationship between aggregate consumption and asset price or aggregate quantity fluctuations that are robust to $1 “mistakes” or misspecifications.”

Theory leaves us with wide priors

Choi [2022] surveys 50 best-selling finance books:

- “There’s basically nothing I saw in the books about how consumption should respond to interest rates.”—suggests households may not pay attention to this, which makes sense given the welfare benefits to doing so.
- “Nine books advise starting with a target for wealth at retirement in order to compute the necessary savings rate”—implies the opposite of substitution.

Conclusion

- Micro literature on how consumption responds to income is now well estabilished.
- Given this evidence, the consumption response to 10 structural shocks is almost entirely explained by the expected path of income.
- This leaves no role for intertemporal substitution
- This is despite some of the structural shocks being accompanied by large and persistent movements in the federal funds rate.

Income and Substitution Effects Decomposition

See Farhi et al. [2022]

Proposition 2. *The response of consumption to a change in the sequence of prices and income $\{dR_s\}$ and $\{dy_s\}$ is given by:*

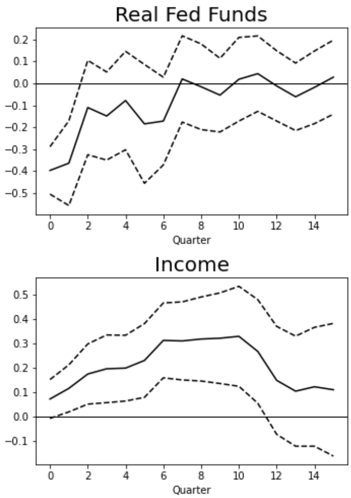
$$\frac{dc_0}{c_0} = -\epsilon(c_0)\mathbb{E}_0^Q \sum_{t=0}^{\tau} \left(\prod_{s=0}^{t} \frac{MPS_s}{R_s} \right) \frac{dR_t}{R_t} + \frac{1}{c_0}\frac{\partial c_0}{\partial a_0}\mathbb{E}_0^{Q^I} \sum_{t=0}^{\tau} \left(\prod_{s=0}^{t-1} \frac{1}{R_s} \right) \left(\frac{a_{t+1}}{R_t}\frac{dR_t}{R_t} + dy_t \right) \quad (1)$$

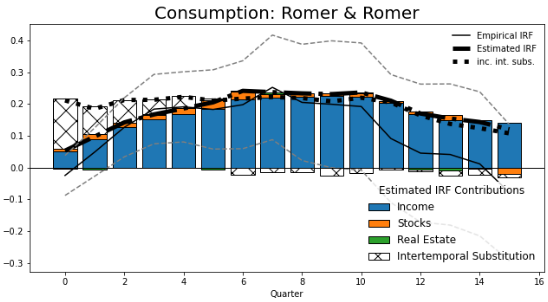
for probability distributions Q and Q^I defined by

$$\frac{dQ}{dP}(\theta^{t+1} \mid \theta^t) \equiv \frac{V'(a_{t+1}, \theta^{t+1})}{\mathbb{E}\left(V'(a_{t+1}, \theta^{t+1}) \mid \theta^t\right)} = \frac{u'(c(a_{t+1}, \theta^{t+1}))}{\mathbb{E}\left(u'(c(a_{t+1}, \theta^{t+1})) \mid \theta^t\right)}, \quad (2)$$

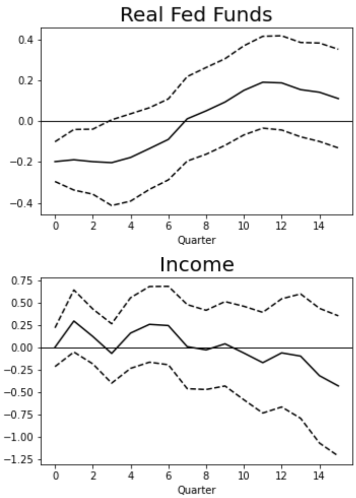
$$\frac{dQ^I}{dP}(\theta^{t+1} \mid \theta^t) \equiv \frac{V''(a_{t+1}, \theta^{t+1})}{\mathbb{E}\left(V''(a_{t+1}, \theta^{t+1}) \mid \theta^t\right)} = \frac{u''\partial_a c(a_{t+1}, \theta^{t+1})}{\mathbb{E}\left(u''\partial_a c(a_{t+1}, \theta^{t+1}) \mid \theta^t\right)}. \quad (3)$$

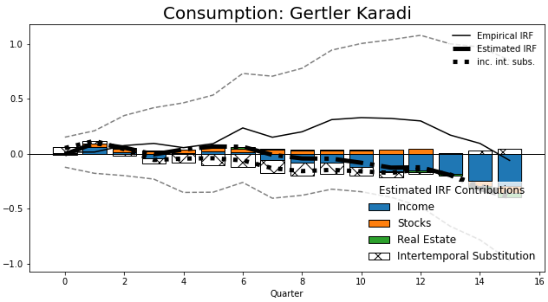
Romer and Romer [2004]



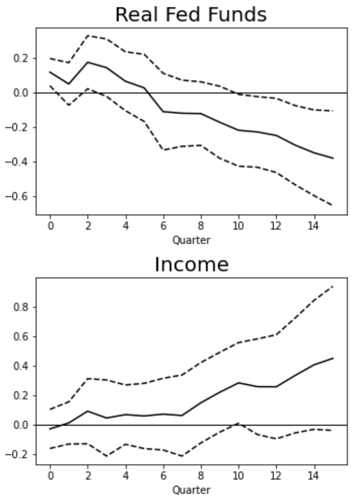


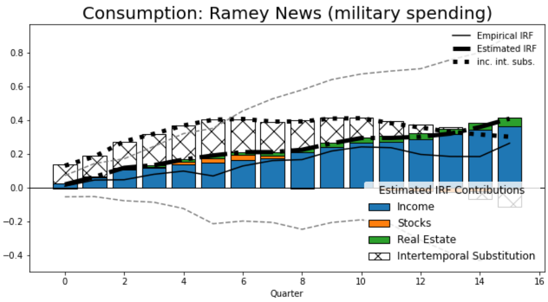
Gertler and Karadi [2015]



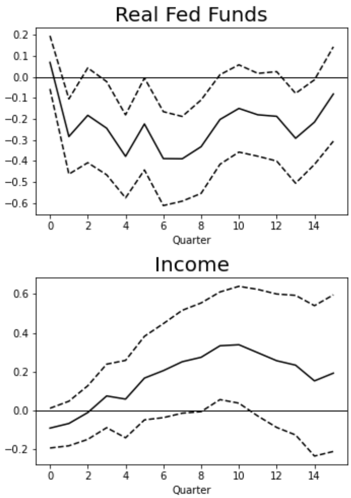


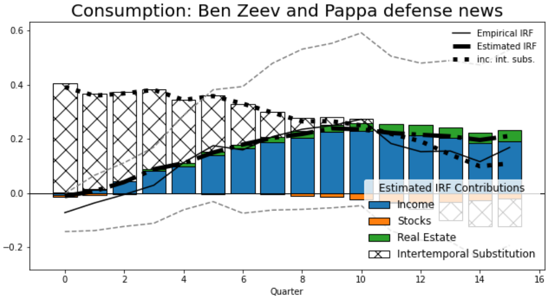
Ramey and Zubairy [2018] military news



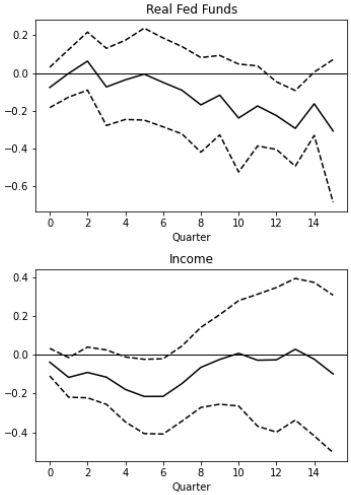


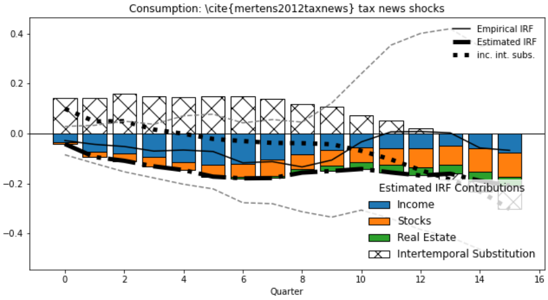
Ben Zeev and Pappa [2017] defence spending shocks



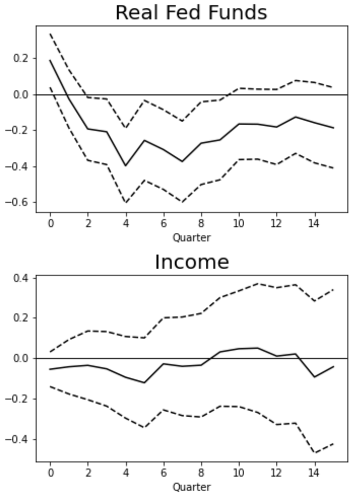


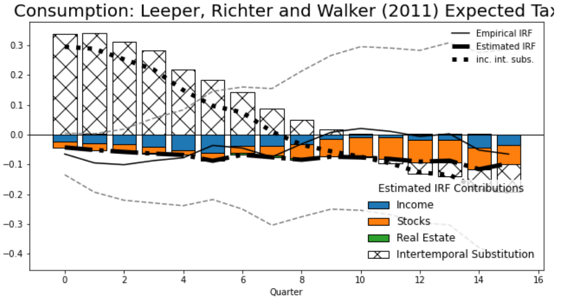
Mertens and Ravn [2012] tax news shocks



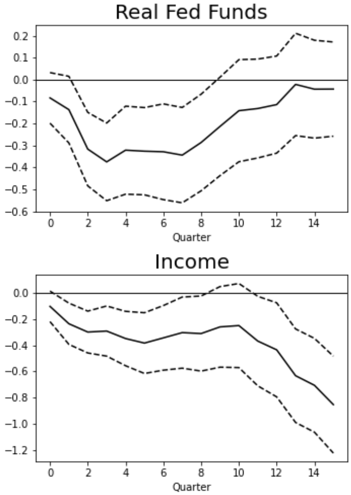


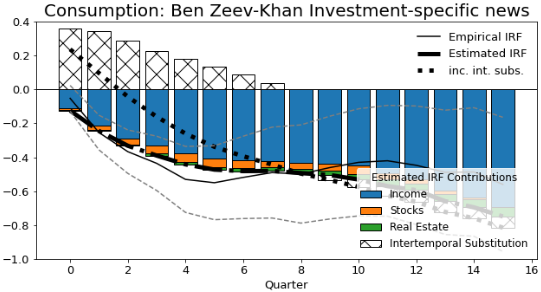
Leeper et al. [2012] expected taxes from one to five years forward



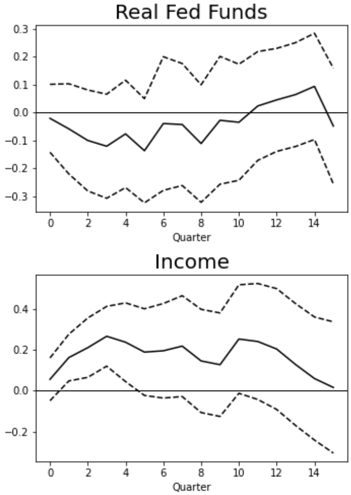


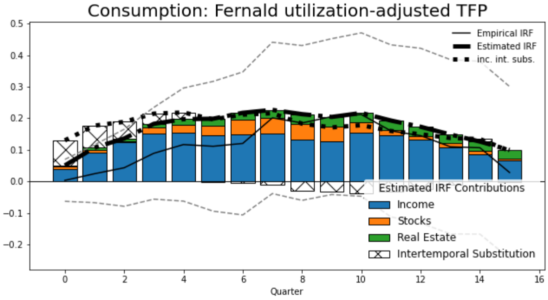
Ben Zeev and Khan [2015] investment specific news shocks



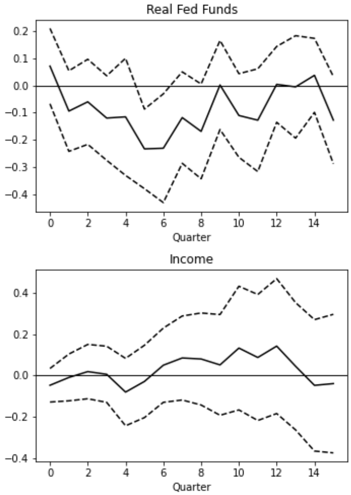


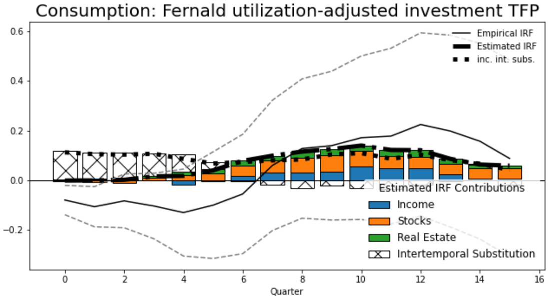
Fernald [2012] utilization-adjusted TFP



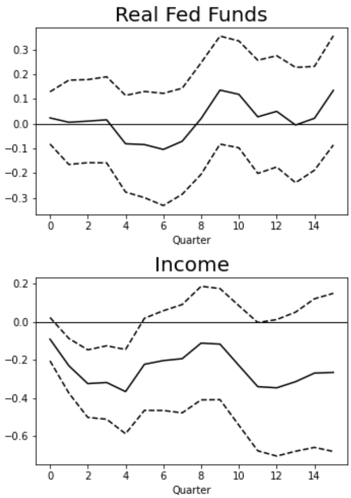


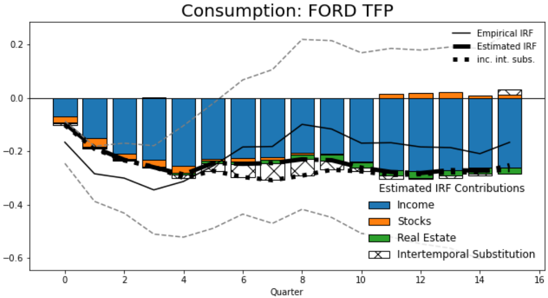
Fernald [2012] utilization-adjusted investment TFP





Francis et al. [2014] unanticipated TFP shocks





Section 5

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